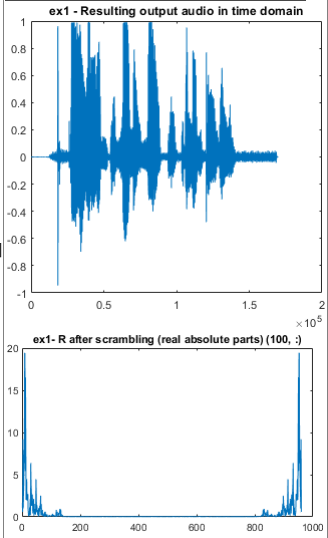
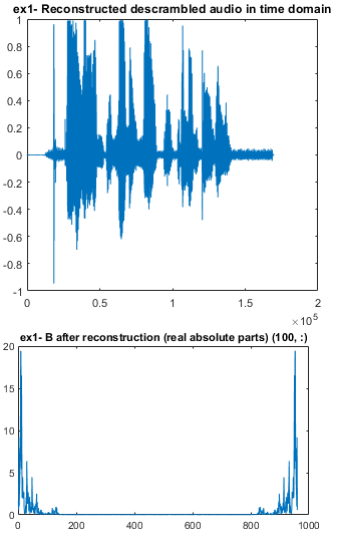
Scrambling Experiment 3

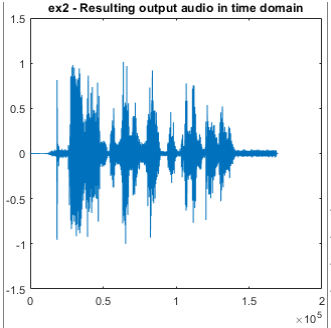
Various Scrambling/Descrambling Configurations

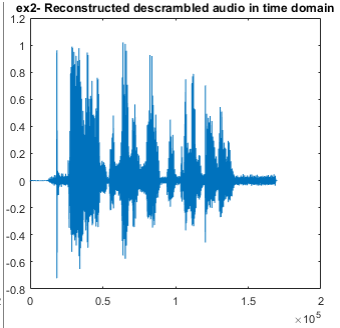
Image resolution for plots: 80

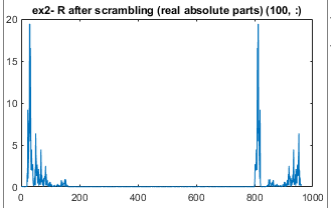
Fs = 48000, Fc = 0.29167

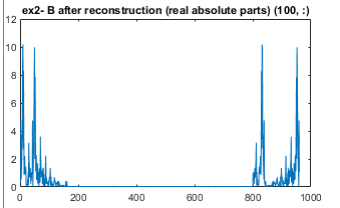
n = 169303, 169920, 169920 for input, scrambled, and reconstructed audio respectively.

Config	Scrambling	Reconstruction
[0 1 2 3 4 5 6 7]	0:00 / 0:04 	0:00 / 0:04 
[7 0 1 2 3 4 5 6]	0:00 / 0:04	0:00 / 0:04

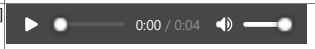

ex2 - Resulting output audio in time domain
$\times 10^5$


ex2- Reconstructed descrambled audio in time domain
$\times 10^5$


ex2- R after scrambling (real absolute parts) (100, :)


ex2- B after reconstruction (real absolute parts) (100, :)

[6 7 0 1 2 3 4 5]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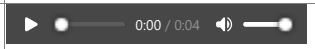
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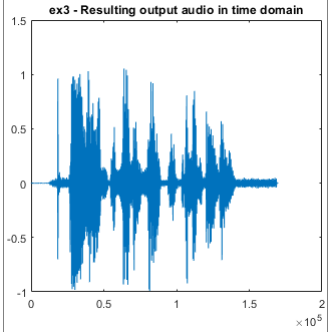
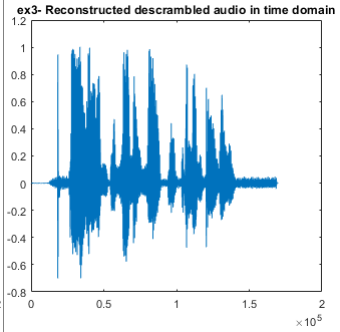
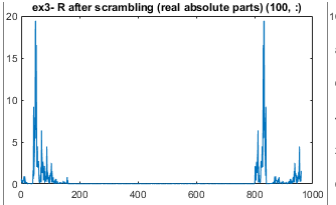
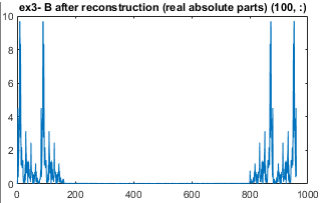
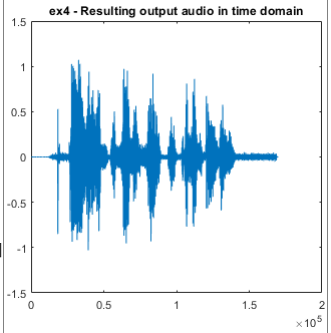
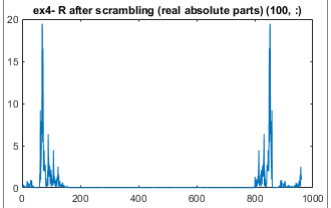
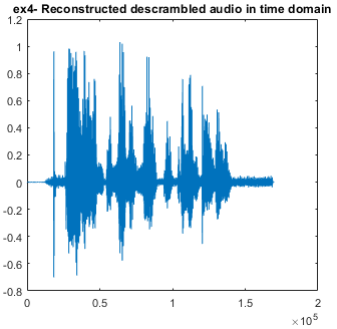
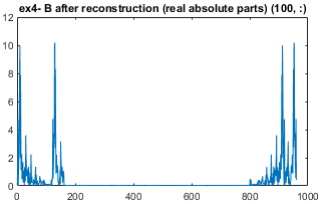
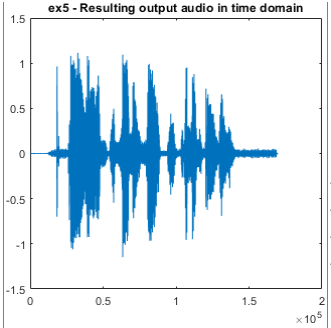
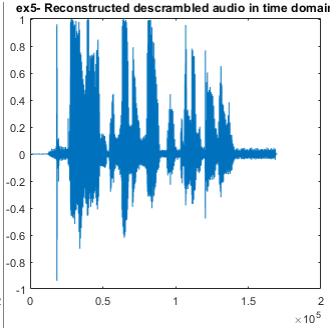
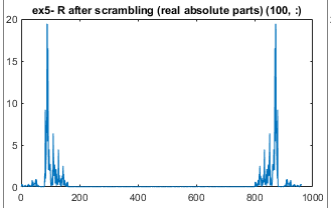

ex3 - Resulting output audio in time domain
$\times 10^5$

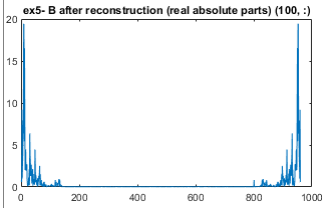

ex3- Reconstructed descrambled audio in time domain
$\times 10^5$

		
[5 6 7 0 1 2 3 4]	0:00 / 0:04  	0:00 / 0:04  
[4 5 6 7 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

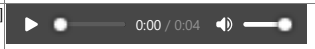

ex5 - Resulting output audio in time domain
×10^5

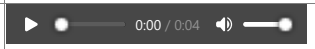

ex5- Reconstructed descrambled audio in time domain
×10^5

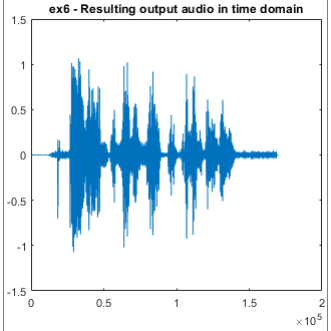

ex5- R after scrambling (real absolute parts) (100, :)

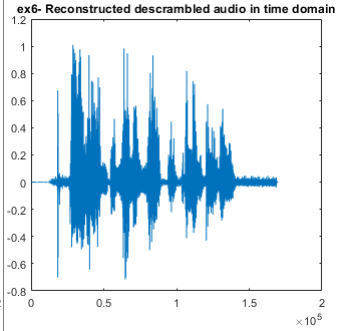

ex5- B after reconstruction (real absolute parts) (100, :)

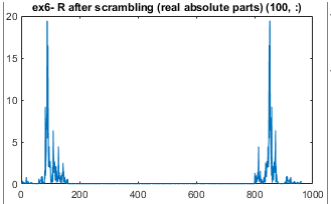
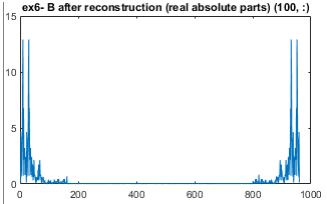
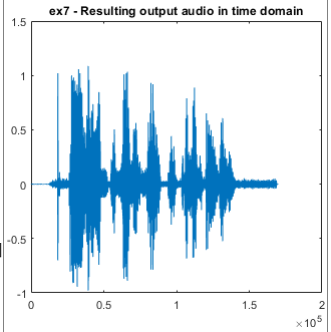
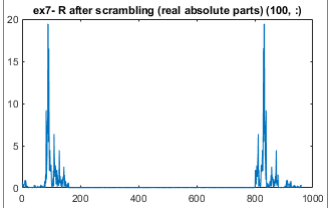
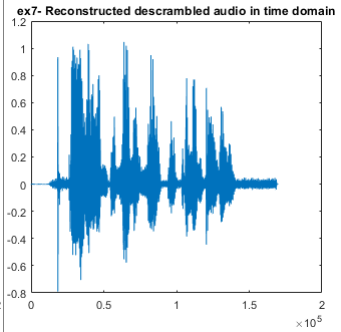
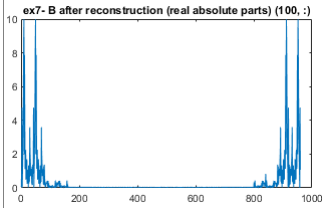
[5 4 7 6 0 1 2 3]

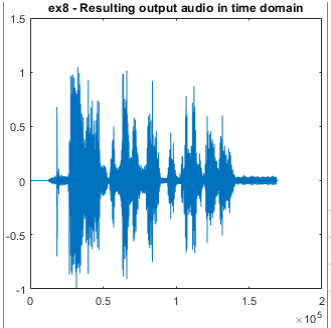
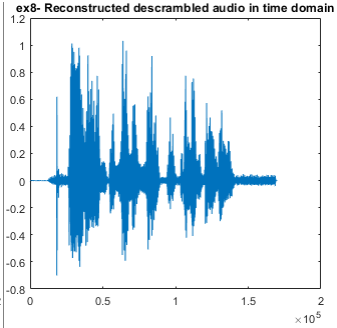
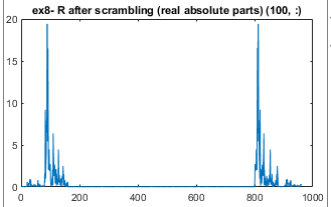
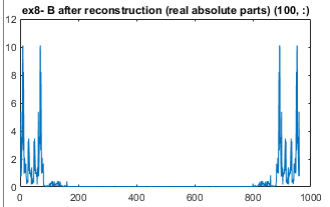
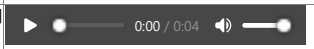
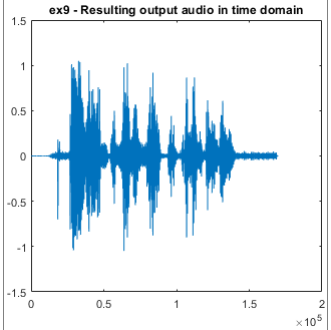
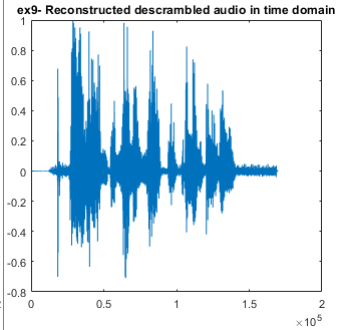

0:00 / 0:04

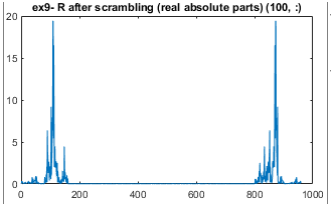
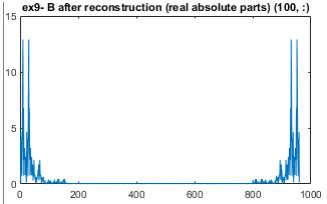
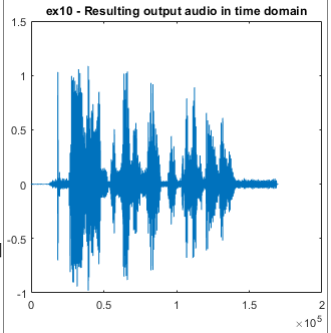
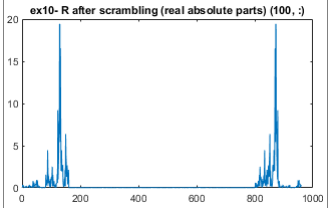
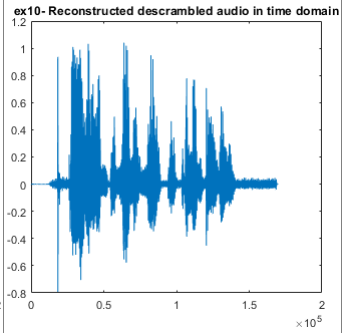
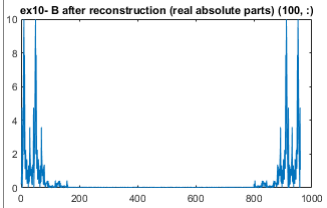

0:00 / 0:04

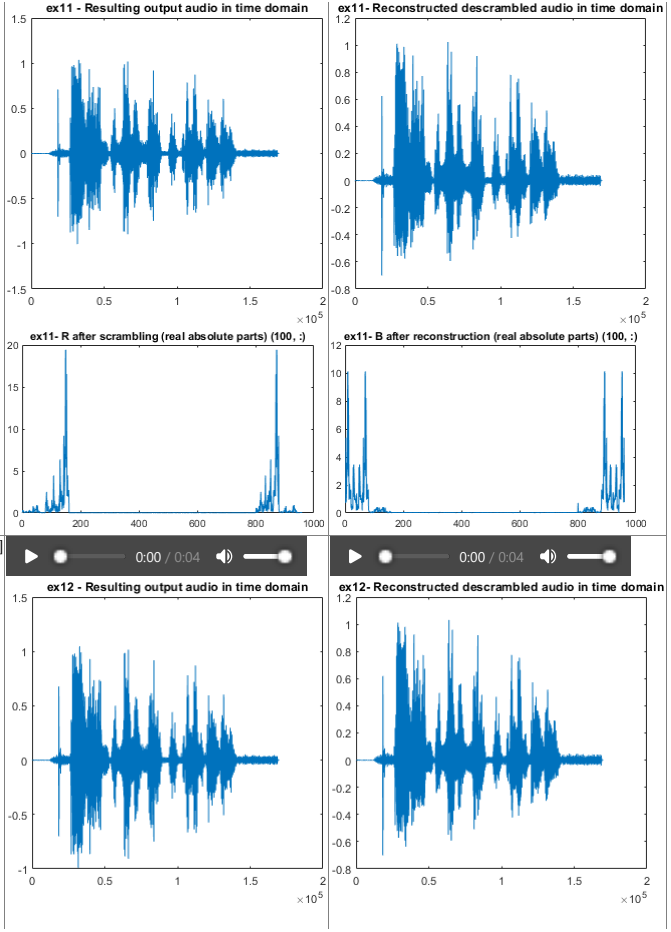

ex6 - Resulting output audio in time domain
×10^5


ex6- Reconstructed descrambled audio in time domain
×10^5

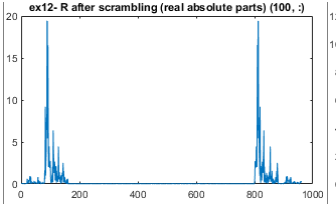
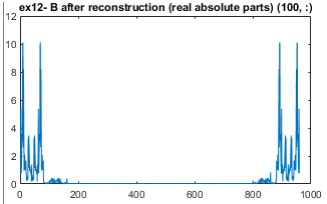
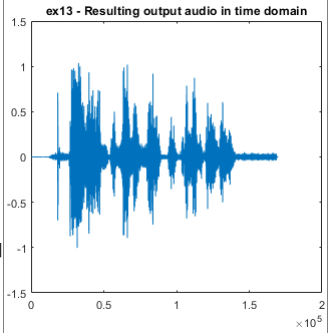
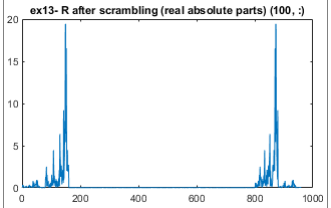
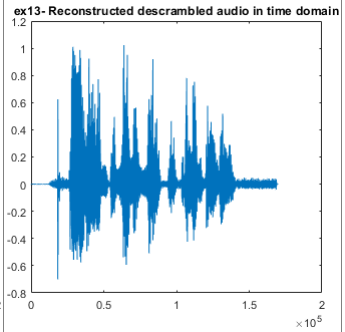
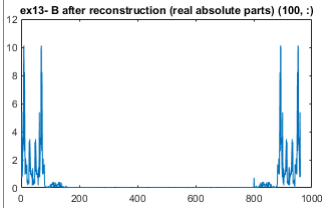
		
[6 7 4 5 0 1 2 3]	0:00 / 0:04  	0:00 / 0:04  
[7 6 5 4 0 1 2 3]	0:00 / 0:04	0:00 / 0:04

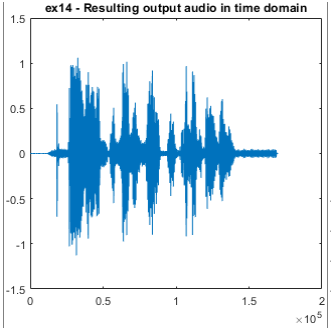
	 ex8 - Resulting output audio in time domain  ex8- Reconstructed descrambled audio in time domain  ex8- R after scrambling (real absolute parts) (100, :)  ex8- B after reconstruction (real absolute parts) (100, :)
[4 5 6 7 1 0 3 2]	 0:00 / 0:04 0:00 / 0:04  ex9 - Resulting output audio in time domain  ex9- Reconstructed descrambled audio in time domain

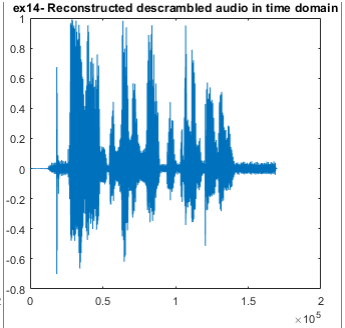
		
[4 5 6 7 2 3 0 1]	0:00 / 0:04  	0:00 / 0:04  
[4 5 6 7 3 2 1 0]	0:00 / 0:04	0:00 / 0:04

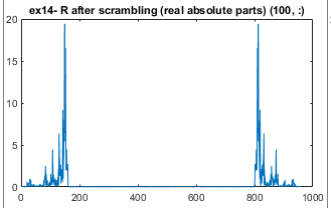

ex11 - Resulting output audio in time domain
ex11- Reconstructed descrambled audio in time domain
ex11- R after scrambling (real absolute parts) (100, :)
ex11- B after reconstruction (real absolute parts) (100, :)
0:00 / 0:04
0:00 / 0:04
ex12 - Resulting output audio in time domain
ex12- Reconstructed descrambled audio in time domain

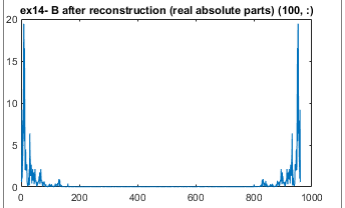
[7 6 5 4 0 1 2 3]

		
[4 5 6 7 3 2 1 0]	0:00 / 0:04  	0:00 / 0:04  
[7 6 4 5 3 2 1 0]	0:00 / 0:04	0:00 / 0:04

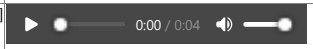

ex14 - Resulting output audio in time domain
×10^5

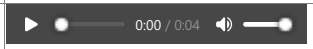

ex14- Reconstructed descrambled audio in time domain
×10^5


ex14- R after scrambling (real absolute parts) (100, :)


ex14- B after reconstruction (real absolute parts) (100, :)

[2 3 0 1 4 5 6 7]


0:00 / 0:04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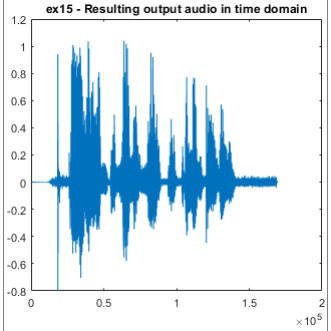
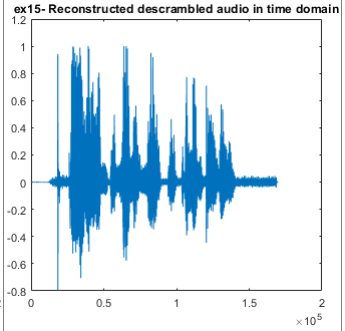
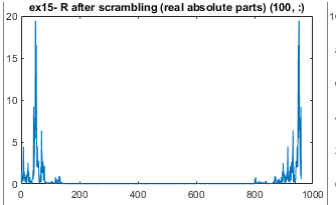
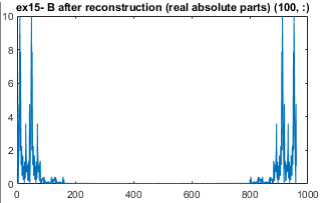
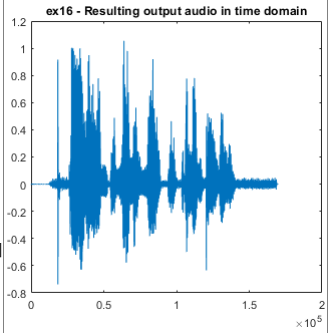
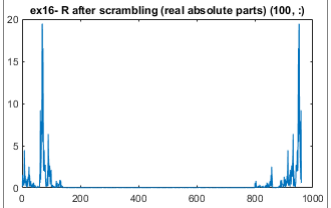
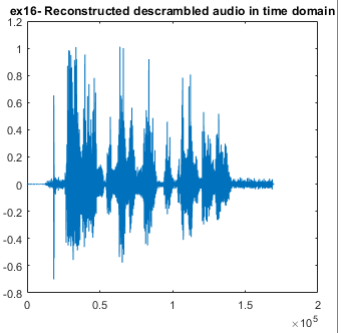
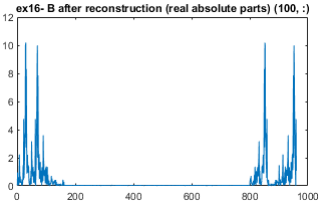
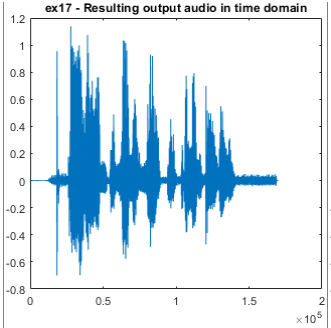
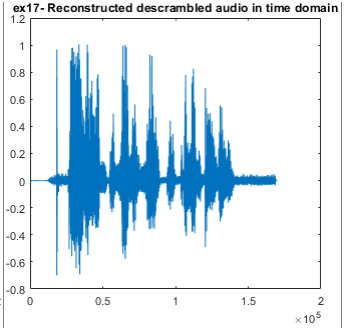
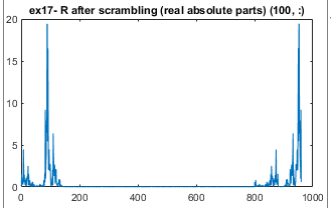

ex15 - Resulting output audio in time domain
×10^5

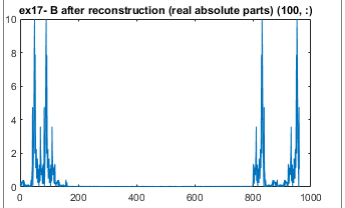

ex15- Reconstructed descrambled audio in time domain
×10^5

		
[2 3 4 0 1 5 6 7]	0:00 / 0:04  	0:00 / 0:04  
[2 3 4 5 0 1 6 7]	0:00 / 0:04	0:00 / 0:04

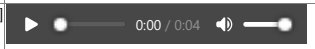

ex17 - Resulting output audio in time domain
$\times 10^5$


ex17- Reconstructed descrambled audio in time domain
$\times 10^5$


ex17- R after scrambling (real absolute parts) (100, :)


ex17- B after reconstruction (real absolute parts) (100, :)

[2 3 4 5 6 0 1 7]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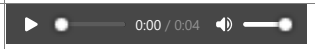
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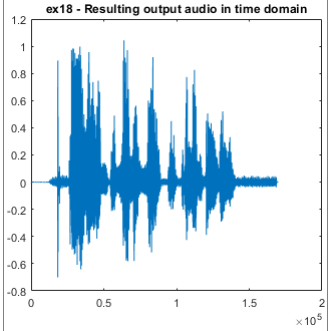
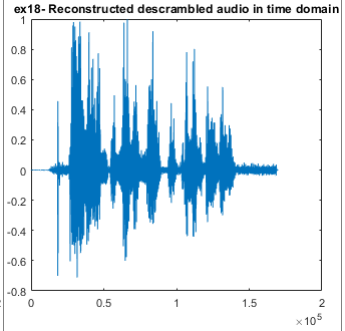
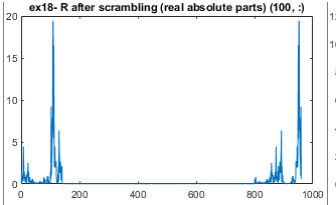
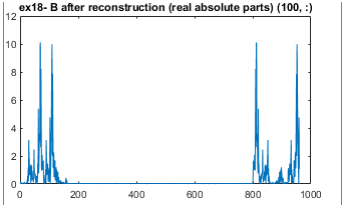
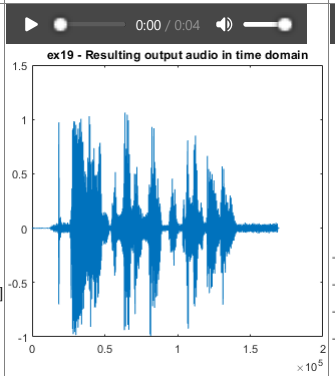
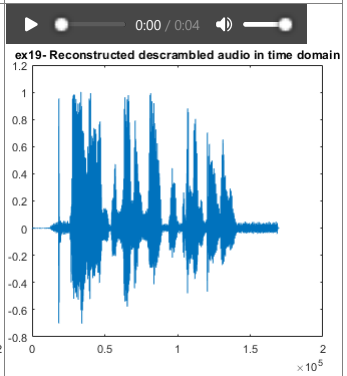

ex18 - Resulting output audio in time domain
$\times 10^5$


ex18- Reconstructed descrambled audio in time domain
$\times 10^5$

	 ex18- R after scrambling (real absolute parts) (100, :)	 ex18- B after reconstruction (real absolute parts) (100, :)
[2 3 4 5 6 7 0 1]	 0:00 / 0:04 ex19 - Resulting output audio in time domain ex19- R after scrambling (real absolute parts) (100, :)	 0:00 / 0:04 ex19- Reconstructed descrambled audio in time domain ex19- B after reconstruction (real absolute parts) (100, :)